

Customer Churn Analysis & Retention Intelligence

OTT SUBSCRIPTION PLATFORM DASHBOARD

Tech Stack: SQL (SQLite), Python (Pandas, NumPy, Matplotlib, Seaborn), Jupyter Notebook

1. Business Context

Subscription-driven platforms live and die by retention. Acquiring a new subscriber costs far more than keeping an existing one, so the ability to identify which customers are at risk of cancelling—and why—is a direct lever on revenue.

This project analyzes a multi-table subscriber dataset (customer demographics, subscription/contract details, and support interactions) for an OTT-style subscription business to answer three questions: who is churning, why they're churning, and where the revenue exposure is concentrated.

2. Approach & Tech Stack

- **Data extraction:** Connected to a relational SQLite database (3 linked tables) using Python's `sqlite3` and `pandas`.
- **Data cleaning:** Type correction, column standardization, missing-value handling, and de-duplication of a one-to-many support table to prevent row fan-out during joins.
- **Feature engineering:** Derived churn flag, customer tenure, complaint counts, and a churn risk tier (Low / Medium / High) from a continuous churn score.
- **Analysis:** Aggregation and group-by analysis across plan type, contract type, state, and support escalation to quantify churn drivers.
- **Visualization:** Matplotlib and Seaborn for trend, comparison, and correlation views.

3. Data Model

The source database (`customer_churn.db`) contains three related tables joined on `customerid`:

Table	Key Fields
db_customer	customerid, name, country, state, gender, dob
db_subscription	customerid, subscription_start_date, plan_type, contract_type, cancellation_date, monthly_charges, cltv, churn_score
db_support	customerid, complaint_date, escalations, csat_score

4. Key Metrics Tracked

KPI	Definition
Churn Rate	Churned customers / total customers
Retention Rate	1 - Churn Rate
ARPU	Average monthly revenue per user
Revenue at Risk	Monthly revenue tied to already-churned customers
Avg. Tenure	Avg days from subscription start to cancellation (or today)
Escalation Rate	Share of support interactions flagged as escalated
Escalation-Churn Correlation	Point-biserial correlation between escalation flag and churn

5. Headline Numbers

28.6%	71.4%	Rs 18.85	1,490 days
CHURN RATE	RETENTION RATE	ARPU	AVG. TENURE
Rs 73.94	18.7%	19.1%	0.47
REVENUE AT RISK	% REVENUE LOST	ESCALATION RATE	ESCALATION-CHURN CORR.

6. Key Insights

6.1 Contract type is the single strongest churn driver

Monthly-contract subscribers churn at **55.6%**, versus **8.3%** for annual-contract subscribers—roughly a **6.7x gap**. Of the 6 churned customers in the dataset, 5 were on monthly contracts. This is the clearest, most actionable signal in the data: contract structure predicts churn far more strongly than plan tier or geography.

6.2 Basic plan carries the highest churn, but limited revenue exposure

The Basic plan churns at **60%**, well above Standard (22.2%) and Premium (14.3%). However, because Basic is the lowest-ARPU tier, this churn is a volume/experience problem rather than a major revenue threat on its own. The bigger dollar exposure sits wherever high-CLTV customers overlap with monthly contracts.

6.3 Support escalations correlate with churn

19.1% of support interactions in the dataset were escalated, and escalation status shows a moderate positive correlation with churn ($r = 0.47$). Customers who escalate are meaningfully more likely to leave; support friction is a leading indicator, not just a downstream symptom.

6.4 Churn is not evenly spread geographically

Karnataka shows 100% churn in this sample (2 of 2 customers), and Meghalaya shows 66.7% (2 of 3). Sample sizes per state are small, so these are directional flags for investigation rather than statistically firm conclusions—but they're worth a root-cause check (pricing changes, service quality, local competition).

6.5 Cancellations cluster in September

Cancellations were concentrated around September 2024, with isolated cancellations in the following months. This kind of clustering is usually tied to a specific trigger—a price change, a service disruption, or a renewal-cycle effect—and is worth cross-referencing against internal change logs.

6.6 Revenue exposure

Churned customers represent Rs 73.94 in monthly recurring revenue (18.7% of total MRR) and Rs 2,047 in lifetime value, out of Rs 17,294 total CLTV in the dataset. Cancellation reasons cited include switching to a competitor, price sensitivity, insufficient content, streaming quality issues, and forgotten free-trial cancellations—a mix of pricing, product, and lifecycle-management problems rather than one single cause.

7. Recommendations

- **Push monthly subscribers toward annual contracts:** This is the single highest-leverage lever available, given the 6.7x churn gap between contract types. A migration incentive (discount, bonus content) targeted at monthly subscribers nearing renewal is the most direct action item.

- **Treat escalations as an early-warning signal:** Customers with recent escalations should be flagged for proactive retention outreach before they cancel, rather than treating escalations purely as a support-quality metric.
- **Investigate the September cancellation spike:** Cross-reference this period against internal records (pricing changes, outages, competitor promotions) to determine whether it was a one-off event or a recurring pattern.
- **Prioritize retention spend by risk tier and CLTV:** Focus efforts by looking at both variables—a Medium-risk, high-CLTV customer is worth more retention effort than a High-risk, low-CLTV one.
- **Scale up state-level analysis:** Re-run the geographic analysis with a larger sample before acting on it operationally; current sample sizes per state (1–4 customers) are too small to be conclusive on their own.

Data Source Note: Synthetic OTT subscriber dataset (21 customers, 3 relational tables) compiled for portfolio and skill-building purposes.